

# Muhammad Hamza

AI & WEB DEVELOPER

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PORTFOLIO [portfolio-lime-rho-96.vercel.app](https://portfolio-lime-rho-96.vercel.app)

## SUMMARY

Computer Science student and developer who ships AI-powered, full-stack products end to end — from React/TypeScript interfaces to Python and Node back ends, with applied work in retrieval-augmented generation (RAG) and LLM-integrated features. I learn by building and deploying real apps.

## SELECTED PROJECTS

### Synapse — AI Study OS

React · FastAPI · RAG · Three.js

A retrieval-augmented “Study OS” that ingests a user’s own documents into a vector store and answers questions grounded in their material, with recalled concepts rendered as an explorable 3D Mind Palace.

[synapse-eight-ruby.vercel.app](https://synapse-eight-ruby.vercel.app) · [github.com/MuhammadHamza2210/Synapse](https://github.com/MuhammadHamza2210/Synapse)

### Orion — Real-Time Stock Dashboard

React · FastAPI · WebSockets · AI

Streams live prices for 50 tickers over WebSockets with interactive charts, plus on-demand LLM analysis that explains any stock’s recent movement in plain language.

[orion-nine-wine.vercel.app](https://orion-nine-wine.vercel.app) · [github.com/MuhammadHamza2210/Orion](https://github.com/MuhammadHamza2210/Orion)

### CodeScan AI — Code Plagiarism Detector

Python · ML · CodeBERT

Combines four engines — AST diffing, a trained ML model, winnowing fingerprints and CodeBERT embeddings — for structure-aware similarity scoring that resists renaming and reformatting.

[github.com/MuhammadHamza2210/CodeScan-AI](https://github.com/MuhammadHamza2210/CodeScan-AI)

**Also built:** LootLo (full-stack e-commerce, Flask/SQLite) · WildMind AI (Gemini-powered encyclopedia) · WeatherScope (Python data viz).

## SKILLS

Languages Python · TypeScript · JavaScript · SQL

AI / ML RAG & LLM apps · Prompt engineering · LLM APIs (Gemini, Claude, Ollama) · embeddings & vector search

Frontend React · Vite · Tailwind CSS · Three.js / R3F · Framer Motion

Backend Node.js · Express · FastAPI · Flask · REST APIs · WebSockets

Tools Git & GitHub · SQLite · Streamlit · Vercel

## EDUCATION

### BS Computer Science — Bahria University, Karachi

2023 – 2027 (expected)

CGPA **3.81 / 4.0** — through 5th semester

Relevant coursework: Data Structures & Algorithms · Databases · OOP · Software Engineering · Operating Systems